ASSESSMENT 2: CONSTRAINT-BASED PROBLEM SOLVING SOLUTIONS

Course Code: CSA17 | Course Name: Artificial Intelligence

1. Doctor Shift Allocation via Backtracking Search

Variables: D1, D2, D3

Domains: {Morning, Afternoon, Night}

Constraints: 1) D1 != Night, 2) D2 must work before D3 (Morning < Afternoon < Night), 3) All doctors assigned unique shifts, 4) D3 != Morning.

- Step 1: Assign D1 = Morning (Valid: D1 != Night).
- Step 2: Assign D2 = Afternoon (Valid: leaves remaining shifts for D3).
- Step 3: Assign D3 = Night (Valid: D3 != Morning, unique shifts, D2 index < D3 index).

Final Valid Schedule: D1 -> Morning, D2 -> Afternoon, D3 -> Night.

2. Robot Grid Navigation with Manhattan Distance (BFS Search)

Problem Constraints: 5x5 grid, uniform move cost = 1, obstacle avoidance, goal-directed navigation.

BFS Execution Details:

- Breadth-First Search (BFS) explores nodes level by level using a FIFO queue. Since edge costs are uniform (cost = 1), BFS guarantees finding the shortest path optimal solution.
- At each step, the robot generates valid neighbor states (Up, Down, Left, Right) while omitting blocked obstacle cells and grid boundaries.
- Node expansion tracks parent pointers to reconstruct the exact optimal coordinate path from Start (S) to Goal (G) once G enters the queue expansion phase.

3. Autonomous Rescue Robot in Disaster Area

- a) Constraint Analysis: Limited sensing requires online updating of the local map; obstacle constraints remove invalid state transitions; risk zone costs (+2) modify standard path costs, forcing the search algorithm to balance distance against safety penalties.
- b) Solution Approach: Online Search Agent / Uniform Cost Search (A*-like heuristic search). The robot senses adjacent cells, updates its internal graph representation dynamically, computes path costs incorporating both movement and risk weights, and steps forward iteratively.
- c) AI Concept Mapping: Intelligent Agent (acts autonomously by perceiving hazards and executing actions); Rationality (maximizes performance with minimum cost and zero safety violations); Environment Type (partially observable, stochastic, sequential, dynamic, discrete).

- d) Justification: An online search strategy is mandatory because the disaster building is partially observable; the robot cannot plan a complete offline path without discovering blocked paths dynamically during traversal.